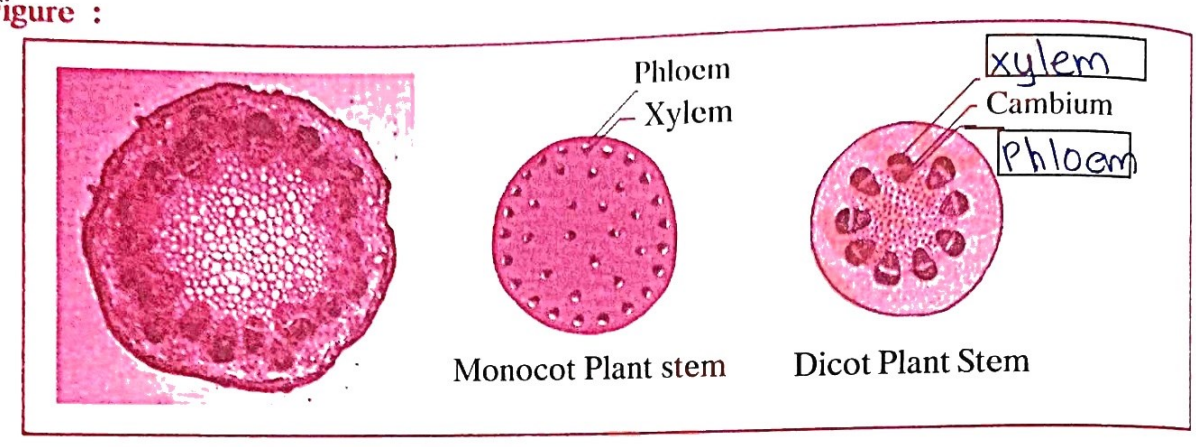
Experiment No. 14

Title : To study the section of stem found in surroundings.

Materials : Tender stem of any dicot and monocot plant which can be cut with blade, Slide, Coverslip, Compound microscope, Petridish, dropper, etc.

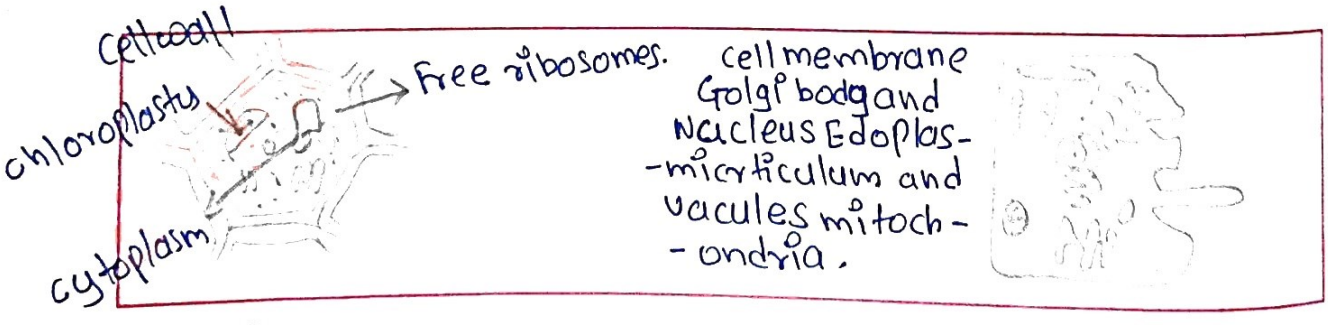
Chemicals : Safranin stain, Glycerine, distilled water, etc.

Figure :



Procedure :

1. Take a tender branchlet of any dicot or monocot plant.
2. Take out 7-8 very thin sections of the branchlet with the help of a new blade.
3. Put those sections in a petridish with dilute solution of safranin for 5 to 7 minutes.
4. Select a good, thin section. Keep it on a slide and put a drop of glycerine on it.
5. Gently and carefully put a coverslip on the section. Observe it under low magnification power of compound microscope.
6. Study the different types of cells and their structures visible in the section.
7. Draw the diagrams of the cells.



Observations :

1. Cells of stem The cell of stem innately undifferentiated cells located in meristem of stem plants.
2. Xylem and Phloem The xylem distributed water. The Phloem carries foods.
3. Tissues The epidermis the ground tissue, and the vascular tissue.

Observation Table :

Features	Name	Functions
Outer Membrane	Epidermis	Protect Body.
Loosely arranged cells	complementary cells	Exchange of gases
Vascular tissue	Xylem	carries water
sieve tube	Phloem	carries food.

Multiple Choice Questions

1. The object is kept at the .....of a slide.
   A. right side B. edge C. left side ✓D. centre
2. A coverslip is kept gently on an object because .....
   A. the object should not get crushed. ✓B. the air bubbles must be avoided.
   C. the coverslip should not break. D. all of above
3. While preparing the slides, stains are used because .....
   A. cell organells should look beautiful ✓B. cell should look clear
   C. cell organells should get magnified D. none of above
4. Xylem and Phloem are .....tissues.
   A. simple permanent ✓B. complex permanent C. meristmatic D. none of above
5. The function of phloem fibre is.....
   ✓A. to conduct food material B. to exchange of gases
   C. to strengthen the plant D. to keep all the cell together

H8HFZG

Questions

1. Explain the difference between xylem and phloem.

1) Xylem consist dead cells, phloem consist living cell.
2) Xylem conduct water and minerals, phloem conduct sugar and amino acid.

2. Describe the variety of structures found in stem cells, according to their functions.

stem cell provide new cell for the body as it grows and replace specialised cells that are damaged or lost.

3. Which tissue is responsible for increasing the breadth and girth of the stem and root?

Apical meristem and lateral meristem tissue responsible for increasing the breadth and girth of the stem and root.